

Assignment 6

Nicole Cruz

2026-08-24

Inhalte

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Figure 2.10: Inequality of Happiness by age group, time and region

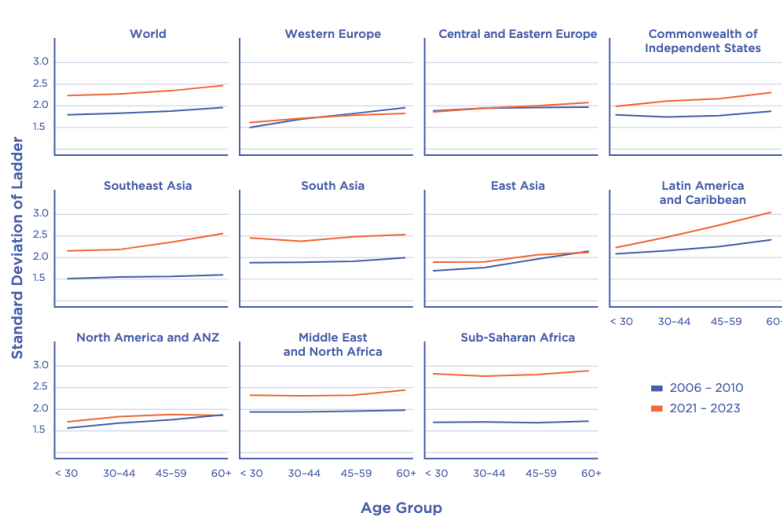


Figure 1: Happiness in different parts of the world over time.

Introduction

Dataset:

A subset of the dataset from the [2024 World Happiness Report](#), which includes the countries for which 15 datapoints or more were available. The report presents the results of a yearly survey of people's happiness in different parts of the world. The initiative is a collaboration of Gallup, the Oxford Wellbeing Research Centre, the UN Sustainable Development Solutions Network, and the WHR's Editorial Board.

The central results are in the first chapters of the report, summarised in its [Figure 2.1](#).

In general, the initiative seeks to highlight happiness as a central indicator of societal progress; that it can be measured; and that its measurement can form an empirical basis for interventions to promote happiness.

In each survey round, around 1000 people from each country are asked several questions related to happiness. In one question they are asked to imagine a ladder with steps from 0 to 10, and to rate their overall life happiness on it, with higher steps representing higher happiness (“Cantril ladder”, corresponding to the variable `Life.Ladder` in the dataset). Further questions refer to positive and negative emotions experienced the past day; and to possible causes of happiness like GDP, social support, freedom to make life choices, and perceived corruption. Below you see the first rows of the dataset.

```
d <- read.csv("countries_n15over.csv")
head(d, 4)

##   Country.name year Life.Ladder Log.GDP.per.capita Social.support
## 1 Afghanistan 2008    3.723590      7.350416      0.4506623
## 2 Afghanistan 2009    4.401778      7.508646      0.5523084
## 3 Afghanistan 2010    4.758381      7.613900      0.5390752
## 4 Afghanistan 2011    3.831719      7.581259      0.5211036
##   Healthy.life.expectancy.at.birth Freedom.to.make.life.choices Generosity
## 1                               50.5                        0.7181143  0.1640551
## 2                               50.8                        0.6788964  0.1872966
## 3                               51.1                        0.6001272  0.1178605
## 4                               51.4                        0.4959014  0.1600984
##   Perceptions.of.corruption Positive.affect Negative.affect
## 1                0.8816863      0.4142970      0.2581955
## 2                0.8500354      0.4814214      0.2370924
## 3                0.7067661      0.5169067      0.2753238
## 4                0.7311085      0.4798347      0.2671747
```

Packages

```
library(kableExtra) # tables
library(ggplot2)    # graphics
library(ggokabeito) # colorblind friendly color palettes
library(brms)       # Bayesian modelling
library(performance) # icc
```

Part 1: Data wrangling

Exercise 1.

Choose 5 countries from a given continent, and create a new, smaller dataset with only the data from those countries (you can use the `subset()` function for this).

Exercise 2.

Formulate a hypothesis to test with your dataset. The hypothesis should be testable using `Life.Ladder` as criterion and one or more of the other variables as predictors.

Exercise 3.

Illustrate the pattern in the data that is relevant to the hypothesis with a graph. Design the graph in such a way, that (a) two (or more) levels of aggregation are visible, and (b) not only information about means, but also information about the variance around those means is visible, (c) the figure axes have comprehensible labels.

Part 2: Data analysis

Exercise 4.

Specify and test a multilevel model to test your hypothesis. Let your model contain at least two levels of aggregation, and include parameters to assess not only means (“fixed effects”) but also variability around those means (“random effects”). Test it both by computing posterior parameter values, and by comparing alternative models (with vs. without a parameter of interest) using Bayes factors.

Part 3: Interpretation

Exercise 5.

Interpret the results against the background of the hypothesis.